

Chapter 22

LP Mailbox

22.1 Overview

ESP32-P4 integrates an LP Mailbox module which provides an efficient inter-core communication mechanism between the LP CPU and HP CPU0/1. The LP Mailbox module comprises of sixteen 32-bit message registers that the LP CPU and HP CPU0/1 can use to store and exchange message. Inter-core communication between LP CPU and HP CPU0/1 is achieved through an interrupt mechanism implemented within the LP Mailbox module.

22.2 Features

ESP32-P4 LP Mailbox has the following features:

- Sixteen 32-bit message registers for inter-core communication
- LP CPU external interrupt signal: [MB_LP_INTR](#)
- HP CPU_x (_x: 0 ~ 1) external interrupt signal: [MB_HP_INTR](#)

22.3 Functional Description

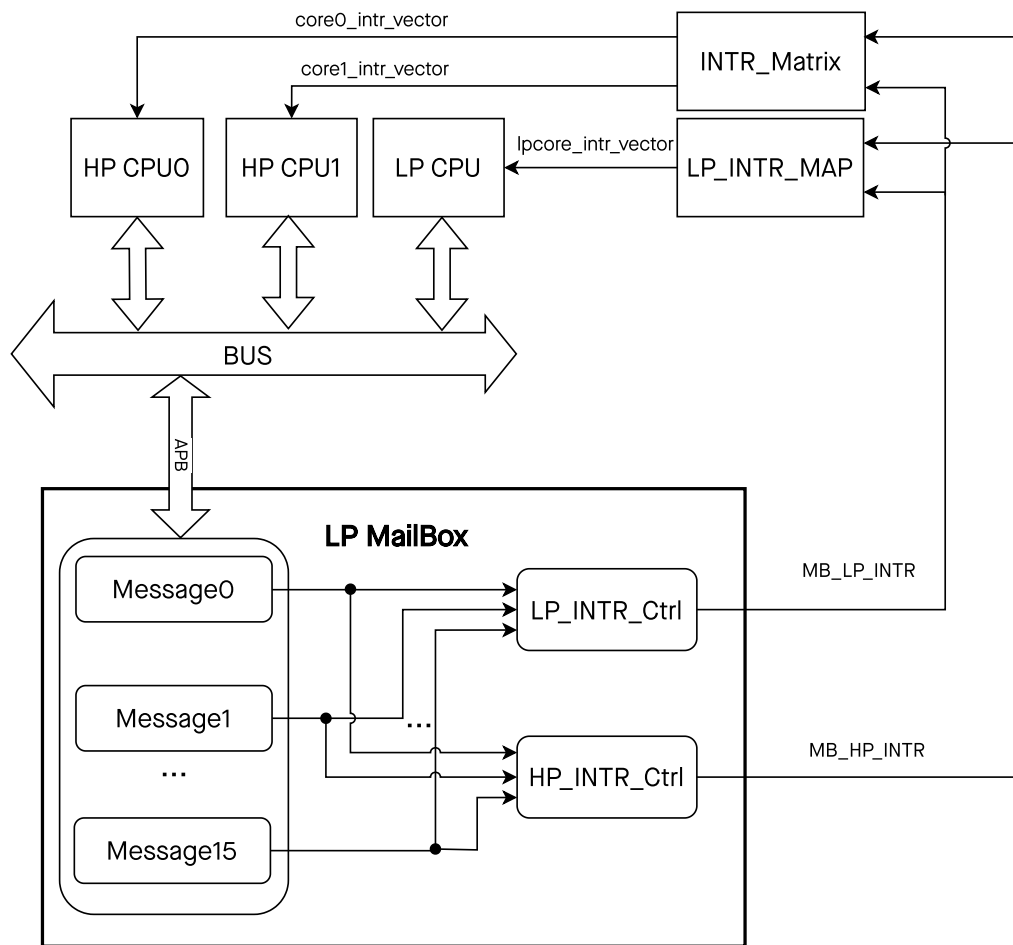


Figure 22.3-1. LP Mailbox Block Diagram

22.3.1 Message Registers

The LP Mailbox provides sixteen 32-bit message registers in total, denoted as [MB_MESSAGE_O_REG](#) ~ [MB_MESSAGE_15_REG](#), for caching inter-core communication messages. Both LP CPU and HP CPU0/1 can read and write these message registers to pass or retrieve message.

Read and write message registers is done through the APB bus, using APB_CLK, which is the only one clock in LP Mailbox. For more information about the clock, please refer to Chapter 10 [Reset and Clock](#).

22.3.2 Interrupts

ESP32-P4 LP Mailbox can generate the following external interrupt signals:

- MB_LP_INTR
- MB_HP_INTR

MB_LP_INTR is directly mapped to LP CPU (see [Low-Power CPU](#)), and MB_HP_INTR can be mapped to HP CPU0/1 through the [Interrupt Matrix](#). These interrupt signals are generated by internal interrupt sources of the

LP Mailbox module. Table 22.3-1 lists the internal interrupt sources, interrupt trigger conditions, and generated interrupt signals of the LP Mailbox.

Table 22.3-1. LP Mailbox's Internal Interrupt Sources

Internal Interrupt Source	Trigger Condition	Interrupt Signal
MB_HP_ <i>m</i> _INT	Write to MB_MESSAGE_ <i>m</i> _REG	MB_HP_INTR
MB_LP_ <i>n</i> _INT	Write to MB_MESSAGE_ <i>n</i> _REG	MB_LP_INTR

Note:

- *m* and *n* can be 0 ~ 15.
- MB_HP_INTR can also serve as an external interrupt source for LP CPU and be mapped to LP CPU by LP_INTR_MAP as illustrated in Figure 22.3-1. For more information about LP_INTR_MAP, please refer to Chapter 3 *Low-Power CPU*.
- MB_LP_INTR can also serve as an external interrupt source for HP CPU_x and can be mapped to HP CPU_x through the *Interrupt Matrix*.
- When the LP Mailbox is required to assign different interrupt sources for HP CPU0 and HP CPU1, it can map MB_HP_INTR and MB_LP_INTR independently to HP CPU0 and HP CPU1 using the *Interrupt Matrix*.

22.3.3 Inter-Core Communication

When communication is required between LP CPU and HP CPU_x, subset of or all message registers in the LP Mailbox can be utilized to share message as needed. The synchronization of message can be achieved through the corresponding interrupts. During communication, software must ensure that only the sender is allowed to write to the message registers to prevent data corruption.

- Example of initiating communication from HP CPU0 to LP CPU using MB_MESSAGE_*n*_REG:
 - Set MB_LP_*n*_INT_ENA to enable the MB_LP_*n*_INT interrupt.
 - HP CPU0 writes to MB_MASSEGE_*n*_REG, caching shared message.
 - LP CPU checks MB_LP_INT_ST_REG interrupt status register, finding the MB_LP_*n*_ST status bit set.
 - LP CPU reads MB_MASSEGE_*n*_REG to retrieve shared message.
- Example of initiating communication from LP CPU to HP CPU0:
 - Set MB_HP_*m*_INT_ENA to enable the MB_HP_*m*_INT interrupt.
 - Configure the *Interrupt Matrix* to map MB_HP_INTR to HP CPU0. See Chapter 12 *Interrupt Matrix*.
 - LP CPU writes to MB_MASSEGE_*m*_REG, caching shared messages.
 - HP CPU0 checks the MB_HP_INT_ST_REG interrupt status register, finding the MB_HP_*m*_ST status bit set.
 - HP CPU0 reads MB_MASSEGE_*m*_REG to retrieve shared messages.

22.4 Register Summary

The addresses in this section are relative to LP Mailbox base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

The abbreviations given in Column **Access** are explained in Section *Access Types for Registers*.

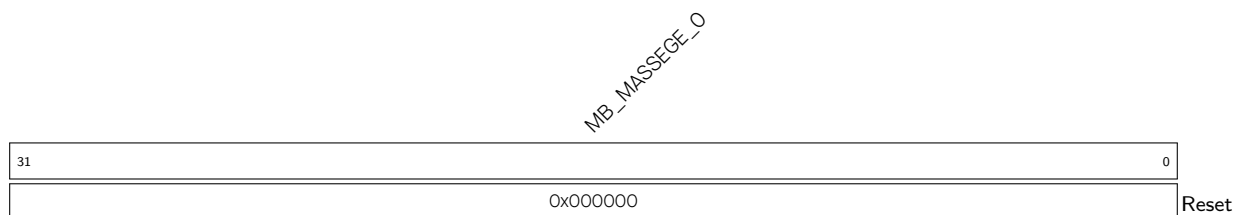
Name	Description	Address	Access
Configuration Registers			
MB_MASSEGE_0_REG	Message register 0	0x0000	R/W
MB_MASSEGE_1_REG	Message register 1	0x0004	R/W
MB_MASSEGE_2_REG	Message register 2	0x0008	R/W
MB_MASSEGE_3_REG	Message register 3	0x000C	R/W
MB_MASSEGE_4_REG	Message register 4	0x0010	R/W
MB_MASSEGE_5_REG	Message register 5	0x0014	R/W
MB_MASSEGE_6_REG	Message register 6	0x0018	R/W
MB_MASSEGE_7_REG	Message register 7	0x001C	R/W
MB_MASSEGE_8_REG	Message register 8	0x0020	R/W
MB_MASSEGE_9_REG	Message register 9	0x0024	R/W
MB_MASSEGE_10_REG	Message register 10	0x0028	R/W
MB_MASSEGE_11_REG	Message register 11	0x002C	R/W
MB_MASSEGE_12_REG	Message register 12	0x0030	R/W
MB_MASSEGE_13_REG	Message register 13	0x0034	R/W
MB_MASSEGE_14_REG	Message register 14	0x0038	R/W
MB_MASSEGE_15_REG	Message register 15	0x003C	R/W
MB_REG_CLK_EN_REG	Clock gate register	0x0060	R/W
Interrupt Registers			
MB_LP_INT_RAW_REG	MB_LP_INTR raw bits	0x0040	RO
MB_LP_INT_ST_REG	MB_LP_INTR status bits	0x0044	RO
MB_LP_INT_ENA_REG	MB_LP_INTR enable bits	0x0048	R/W
MB_LP_INT_CLR_REG	MB_LP_INTR clear bits	0x004C	WO
MB_HP_INT_RAW_REG	MB_HP_INTR raw bits	0x0050	RO
MB_HP_INT_ST_REG	MB_HP_INTR status bits	0x0054	RO
MB_HP_INT_ENA_REG	MB_HP_INTR enable bits	0x0058	R/W
MB_HP_INT_CLR_REG	MB_HP_INTR clear bits	0x005C	WO

22.5 Registers

The addresses in this section are relative to LP Mailbox base address provided in Table 7.3-2 in Chapter 7 *System and Memory*.

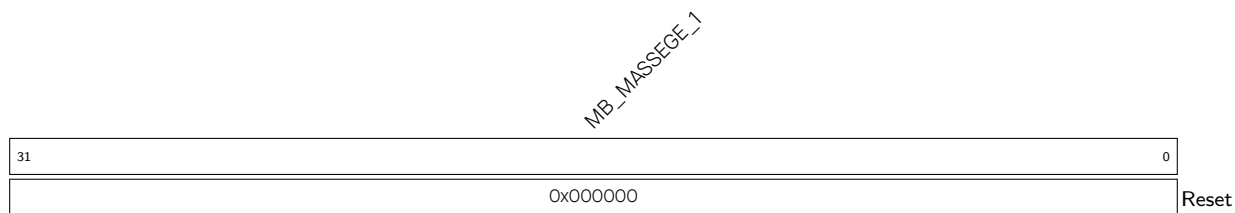
For how to program reserved fields, please refer to Section *Programming Reserved Register Field*.

Register 22.1. MB_MASSEGE_O_REG (0x0000)



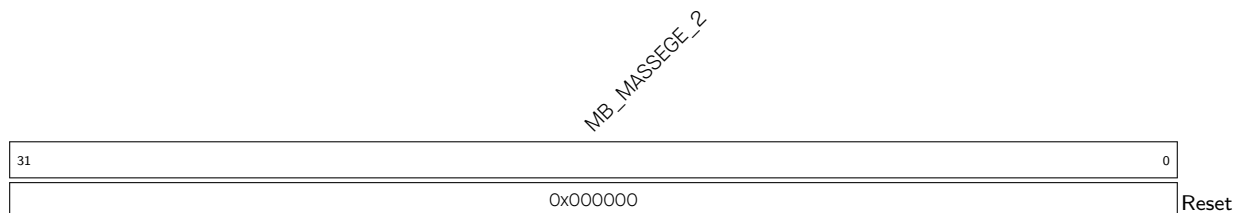
MB_MASSEGE_O Message register 0. (R/W)

Register 22.2. MB_MASSEGE_1_REG (0x0004)



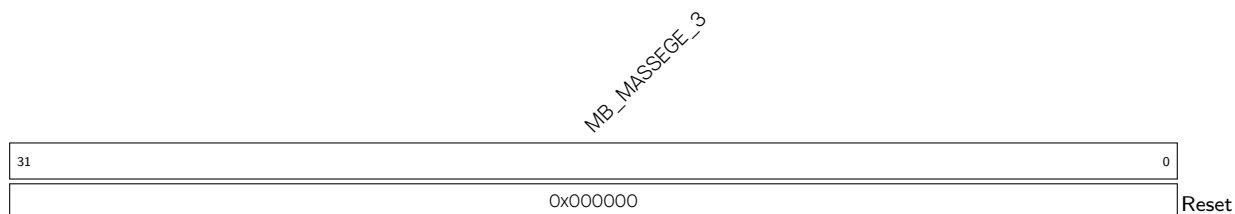
MB_MASSEGE_1 Message register 1. (R/W)

Register 22.3. MB_MASSEGE_2_REG (0x0008)



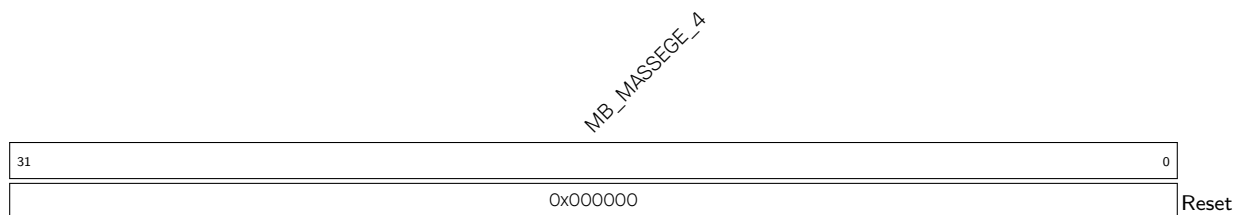
MB_MASSEGE_2 Message register 2. (R/W)

Register 22.4. MB_MASSEGE_3_REG (0x000C)



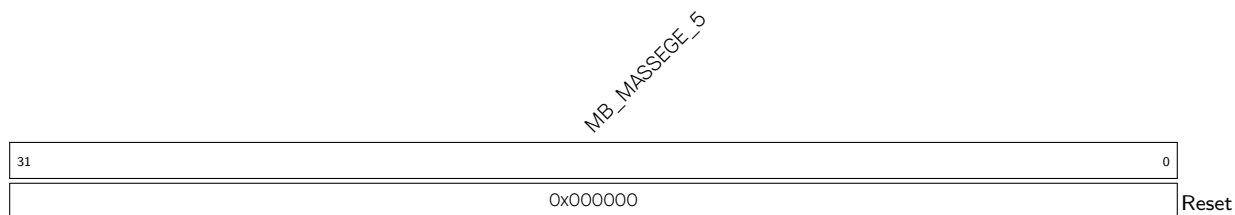
MB_MASSEGE_3 Message register 3. (R/W)

Register 22.5. MB_MASSEGE_4_REG (0x0010)



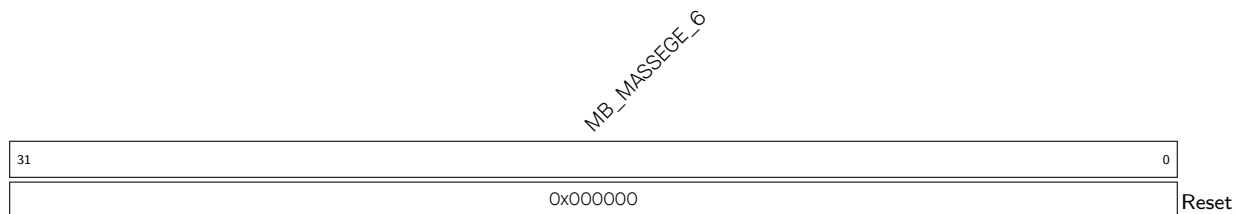
MB_MASSEGE_4 Message register 4. (R/W)

Register 22.6. MB_MASSEGE_5_REG (0x0014)



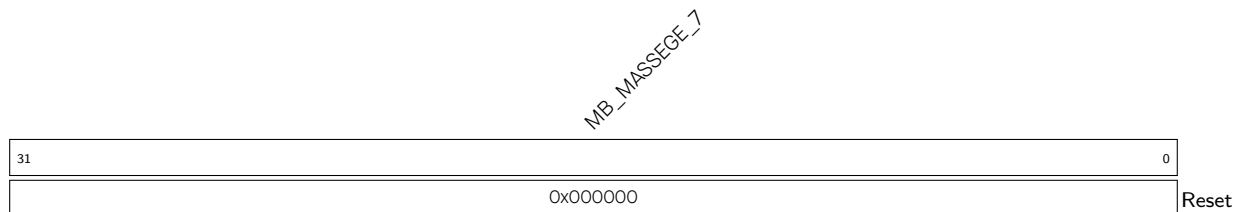
MB_MASSEGE_5 Message register 5. (R/W)

Register 22.7. MB_MASSEGE_6_REG (0x0018)

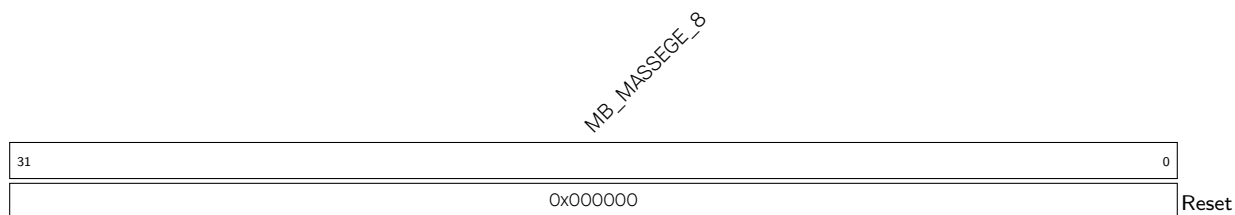
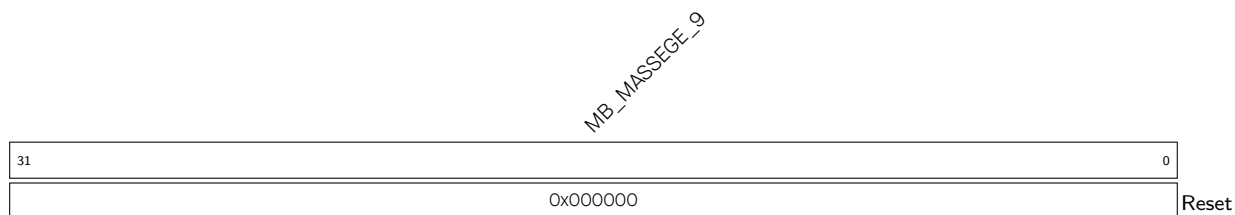
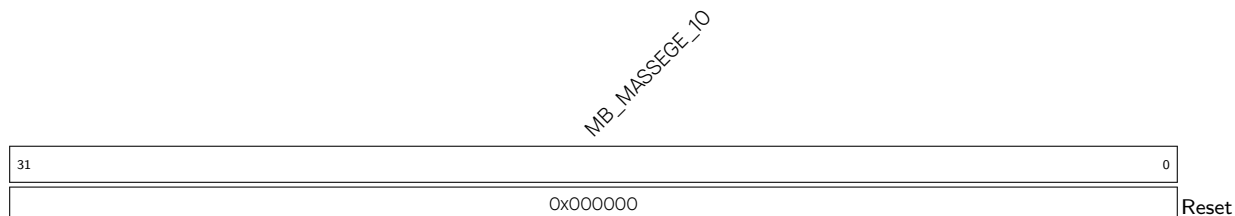
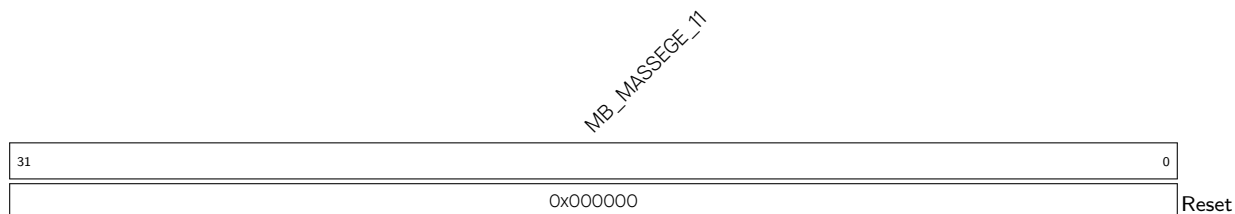


MB_MASSEGE_6 Message register 6. (R/W)

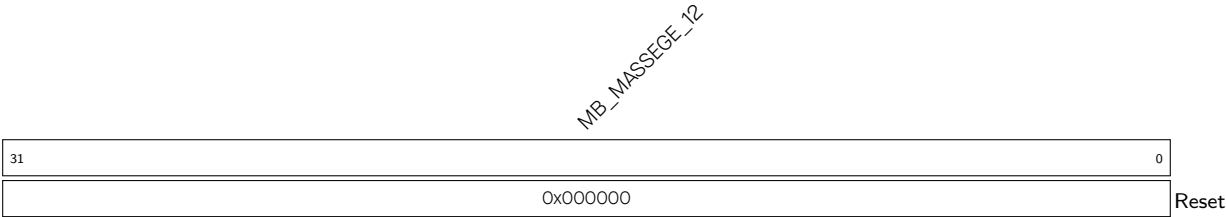
Register 22.8. MB_MASSEGE_7_REG (0x001C)



MB_MASSEGE_7 Message register 7. (R/W)

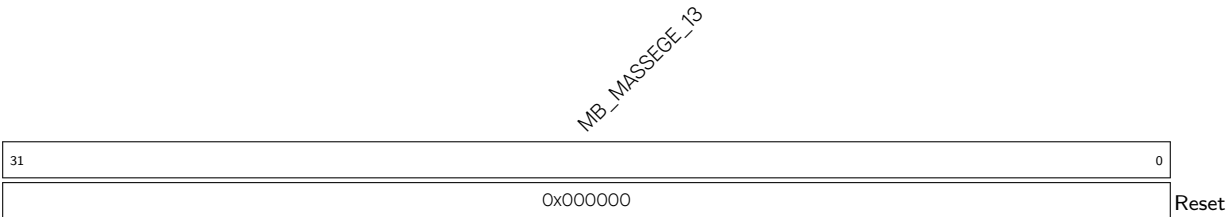
Register 22.9. MB_MASSEGE_8_REG (0x0020)**MB_MASSEGE_8** Message register 8. (R/W)**Register 22.10. MB_MASSEGE_9_REG (0x0024)****MB_MASSEGE_9** Message register 9. (R/W)**Register 22.11. MB_MASSEGE_10_REG (0x0028)****MB_MASSEGE_10** Message register 10. (R/W)**Register 22.12. MB_MASSEGE_11_REG (0x002C)****MB_MASSEGE_11** Message register 11. (R/W)

Register 22.13. MB_MASSEGE_12_REG (0x0030)



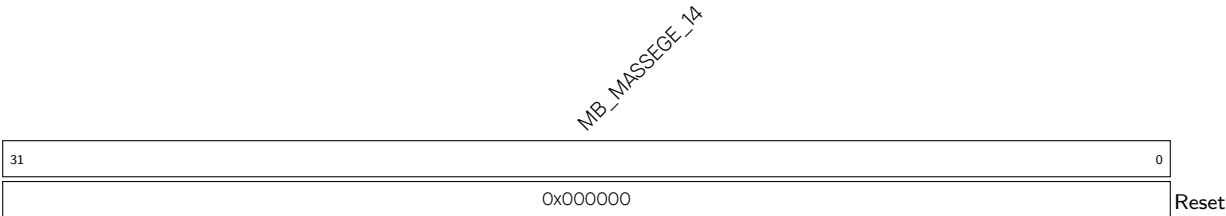
MB_MASSEGE_12 Message register 12. (R/W)

Register 22.14. MB_MASSEGE_13_REG (0x0034)



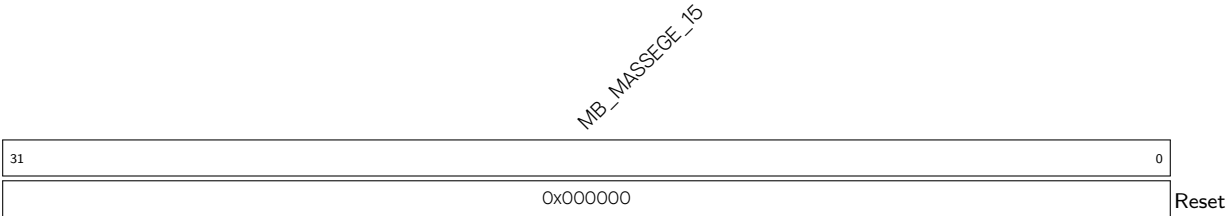
MB_MASSEGE_13 Message register 13. (R/W)

Register 22.15. MB_MASSEGE_14_REG (0x0038)



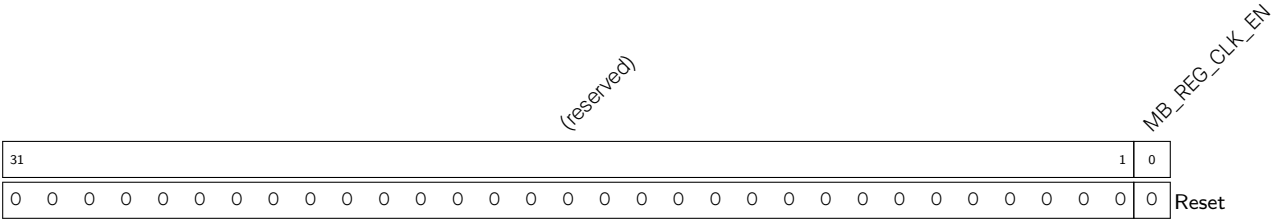
MB_MASSEGE_14 Message register 14. (R/W)

Register 22.16. MB_MASSEGE_15_REG (0x003C)



MB_MASSEGE_15 Message register 15. (R/W)

Register 22.17. MB_REG_CLK_EN_REG (0x0060)



MB_REG_CLK_EN Configures register clock gating.

- 1: Register clock is always enabled for read and write operations.
- 0: Only enable clock for register read or write operations.

(R/W)

Register 22.18. MB_LP_INT_RAW_REG (0x0040)

(reserved)																MB_LP_15_INT_RAW																	
																MB_LP_14_INT_RAW																	
																MB_LP_13_INT_RAW																	
																MB_LP_12_INT_RAW																	
																MB_LP_11_INT_RAW																	
																MB_LP_10_INT_RAW																	
																MB_LP_9_INT_RAW																	
																MB_LP_8_INT_RAW																	
																MB_LP_7_INT_RAW																	
																MB_LP_6_INT_RAW																	
																MB_LP_5_INT_RAW																	
																MB_LP_4_INT_RAW																	
																MB_LP_3_INT_RAW																	
																MB_LP_2_INT_RAW																	
																MB_LP_1_INT_RAW																	
																MB_LP_0_INT_RAW																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						

MB_LP_0_INT_RAW The raw interrupt bit of the MB_LP_0_INT interrupt. (RO)

MB_LP_1_INT_RAW The raw interrupt bit of the MB_LP_1_INT interrupt. (RO)

MB_LP_2_INT_RAW The raw interrupt bit of the MB_LP_2_INT interrupt. (RO)

MB_LP_3_INT_RAW The raw interrupt bit of the MB_LP_3_INT interrupt. (RO)

MB_LP_4_INT_RAW The raw interrupt bit of the MB_LP_4_INT interrupt. (RO)

MB_LP_5_INT_RAW The raw interrupt bit of the MB_LP_5_INT interrupt. (RO)

MB_LP_6_INT_RAW The raw interrupt bit of the MB_LP_6_INT interrupt. (RO)

MB_LP_7_INT_RAW The raw interrupt bit of the MB_LP_7_INT interrupt. (RO)

MB_LP_8_INT_RAW The raw interrupt bit of the MB_LP_8_INT interrupt. (RO)

MB_LP_9_INT_RAW The raw interrupt bit of the MB_LP_9_INT interrupt. (RO)

MB_LP_10_INT_RAW The raw interrupt bit of the MB_LP_10_INT interrupt. (RO)

MB_LP_11_INT_RAW The raw interrupt bit of the MB_LP_11_INT interrupt. (RO)

MB_LP_12_INT_RAW The raw interrupt bit of the MB_LP_12_INT interrupt. (RO)

MB_LP_13_INT_RAW The raw interrupt bit of the MB_LP_13_INT interrupt. (RO)

MB_LP_14_INT_RAW The raw interrupt bit of the MB_LP_14_INT interrupt. (RO)

MB_LP_15_INT_RAW The raw interrupt bit of the MB_LP_15_INT interrupt. (RO)

Register 22.19. MB_LP_INT_ST_REG (0x0044)

(reserved)																MB_LP_15_INT_ST MB_LP_14_INT_ST MB_LP_13_INT_ST MB_LP_12_INT_ST MB_LP_11_INT_ST MB_LP_10_INT_ST MB_LP_9_INT_ST MB_LP_8_INT_ST MB_LP_7_INT_ST MB_LP_6_INT_ST MB_LP_5_INT_ST MB_LP_4_INT_ST MB_LP_3_INT_ST MB_LP_2_INT_ST MB_LP_1_INT_ST MB_LP_0_INT_ST																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

MB_LP_0_INT_ST The masked interrupt status bit of the MB_LP_0_INT interrupt. (RO)

MB_LP_1_INT_ST The masked interrupt status bit of the MB_LP_1_INT interrupt. (RO)

MB_LP_2_INT_ST The masked interrupt status bit of the MB_LP_2_INT interrupt. (RO)

MB_LP_3_INT_ST The masked interrupt status bit of the MB_LP_3_INT interrupt. (RO)

MB_LP_4_INT_ST The masked interrupt status bit of the MB_LP_4_INT interrupt. (RO)

MB_LP_5_INT_ST The masked interrupt status bit of the MB_LP_5_INT interrupt. (RO)

MB_LP_6_INT_ST The masked interrupt status bit of the MB_LP_6_INT interrupt. (RO)

MB_LP_7_INT_ST The masked interrupt status bit of the MB_LP_7_INT interrupt. (RO)

MB_LP_8_INT_ST The masked interrupt status bit of the MB_LP_8_INT interrupt. (RO)

MB_LP_9_INT_ST The masked interrupt status bit of the MB_LP_9_INT interrupt. (RO)

MB_LP_10_INT_ST The masked interrupt status bit of the MB_LP_10_INT interrupt. (RO)

MB_LP_11_INT_ST The masked interrupt status bit of the MB_LP_11_INT interrupt. (RO)

MB_LP_12_INT_ST The masked interrupt status bit of the MB_LP_12_INT interrupt. (RO)

MB_LP_13_INT_ST The masked interrupt status bit of the MB_LP_13_INT interrupt. (RO)

MB_LP_14_INT_ST The masked interrupt status bit of the MB_LP_14_INT interrupt. (RO)

MB_LP_15_INT_ST The masked interrupt status bit of the MB_LP_15_INT interrupt. (RO)

Register 22.20. MB_LP_INT_ENA_REG (0x0048)

(reserved)																MB_LP_15_INT_ENA MB_LP_14_INT_ENA MB_LP_13_INT_ENA MB_LP_12_INT_ENA MB_LP_11_INT_ENA MB_LP_10_INT_ENA MB_LP_9_INT_ENA MB_LP_8_INT_ENA MB_LP_7_INT_ENA MB_LP_6_INT_ENA MB_LP_5_INT_ENA MB_LP_4_INT_ENA MB_LP_3_INT_ENA MB_LP_2_INT_ENA MB_LP_1_INT_ENA MB_LP_0_INT_ENA																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	Reset							

MB_LP_0_INT_ENA Write 1 to enable the MB_LP_0_INT interrupt. (R/W)

MB_LP_1_INT_ENA Write 1 to enable the MB_LP_1_INT interrupt. (R/W)

MB_LP_2_INT_ENA Write 1 to enable the MB_LP_2_INT interrupt. (R/W)

MB_LP_3_INT_ENA Write 1 to enable the MB_LP_3_INT interrupt. (R/W)

MB_LP_4_INT_ENA Write 1 to enable the MB_LP_4_INT interrupt. (R/W)

MB_LP_5_INT_ENA Write 1 to enable the MB_LP_5_INT interrupt. (R/W)

MB_LP_6_INT_ENA Write 1 to enable the MB_LP_6_INT interrupt. (R/W)

MB_LP_7_INT_ENA Write 1 to enable the MB_LP_7_INT interrupt. (R/W)

MB_LP_8_INT_ENA Write 1 to enable the MB_LP_8_INT interrupt. (R/W)

MB_LP_9_INT_ENA Write 1 to enable the MB_LP_9_INT interrupt. (R/W)

MB_LP_10_INT_ENA Write 1 to enable the MB_LP_10_INT interrupt. (R/W)

MB_LP_11_INT_ENA Write 1 to enable the MB_LP_11_INT interrupt. (R/W)

MB_LP_12_INT_ENA Write 1 to enable the MB_LP_12_INT interrupt. (R/W)

MB_LP_13_INT_ENA Write 1 to enable the MB_LP_13_INT interrupt. (R/W)

MB_LP_14_INT_ENA Write 1 to enable the MB_LP_14_INT interrupt. (R/W)

MB_LP_15_INT_ENA Write 1 to enable the MB_LP_15_INT interrupt. (R/W)

Register 22.21. MB_LP_INT_CLR_REG (0x004C)

(reserved)																MB_LP_15_INT_CLR																	
																MB_LP_14_INT_CLR																	
																MB_LP_13_INT_CLR																	
																MB_LP_12_INT_CLR																	
																MB_LP_11_INT_CLR																	
																MB_LP_10_INT_CLR																	
																MB_LP_9_INT_CLR																	
																MB_LP_8_INT_CLR																	
																MB_LP_7_INT_CLR																	
																MB_LP_6_INT_CLR																	
																MB_LP_5_INT_CLR																	
																MB_LP_4_INT_CLR																	
																MB_LP_3_INT_CLR																	
																MB_LP_2_INT_CLR																	
																MB_LP_1_INT_CLR																	
																MB_LP_0_INT_CLR																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						

MB_LP_0_INT_CLR Write 1 to clear the MB_LP_0_INT interrupt. (WO)

MB_LP_1_INT_CLR Write 1 to clear the MB_LP_1_INT interrupt. (WO)

MB_LP_2_INT_CLR Write 1 to clear the MB_LP_2_INT interrupt. (WO)

MB_LP_3_INT_CLR Write 1 to clear the MB_LP_3_INT interrupt. (WO)

MB_LP_4_INT_CLR Write 1 to clear the MB_LP_4_INT interrupt. (WO)

MB_LP_5_INT_CLR Write 1 to clear the MB_LP_5_INT interrupt. (WO)

MB_LP_6_INT_CLR Write 1 to clear the MB_LP_6_INT interrupt. (WO)

MB_LP_7_INT_CLR Write 1 to clear the MB_LP_7_INT interrupt. (WO)

MB_LP_8_INT_CLR Write 1 to clear the MB_LP_8_INT interrupt. (WO)

MB_LP_9_INT_CLR Write 1 to clear the MB_LP_9_INT interrupt. (WO)

MB_LP_10_INT_CLR Write 1 to clear the MB_LP_10_INT interrupt. (WO)

MB_LP_11_INT_CLR Write 1 to clear the MB_LP_11_INT interrupt. (WO)

MB_LP_12_INT_CLR Write 1 to clear the MB_LP_12_INT interrupt. (WO)

MB_LP_13_INT_CLR Write 1 to clear the MB_LP_13_INT interrupt. (WO)

MB_LP_14_INT_CLR Write 1 to clear the MB_LP_14_INT interrupt. (WO)

MB_LP_15_INT_CLR Write 1 to clear the MB_LP_15_INT interrupt. (WO)

Register 22.22. MB_HP_INT_RAW_REG (0x0050)

(reserved)																MB_HP_15_INT_RAW MB_HP_14_INT_RAW MB_HP_13_INT_RAW MB_HP_12_INT_RAW MB_HP_11_INT_RAW MB_HP_10_INT_RAW MB_HP_9_INT_RAW MB_HP_8_INT_RAW MB_HP_7_INT_RAW MB_HP_6_INT_RAW MB_HP_5_INT_RAW MB_HP_4_INT_RAW MB_HP_3_INT_RAW MB_HP_2_INT_RAW MB_HP_1_INT_RAW MB_HP_0_INT_RAW																
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	Reset	

MB_HP_0_INT_RAW The raw interrupt bit of the MB_HP_0_INT interrupt. (RO)

MB_HP_1_INT_RAW The raw interrupt bit of the MB_HP_1_INT interrupt. (RO)

MB_HP_2_INT_RAW The raw interrupt bit of the MB_HP_2_INT interrupt. (RO)

MB_HP_3_INT_RAW The raw interrupt bit of the MB_HP_3_INT interrupt. (RO)

MB_HP_4_INT_RAW The raw interrupt bit of the MB_HP_4_INT interrupt. (RO)

MB_HP_5_INT_RAW The raw interrupt bit of the MB_HP_5_INT interrupt. (RO)

MB_HP_6_INT_RAW The raw interrupt bit of the MB_HP_6_INT interrupt. (RO)

MB_HP_7_INT_RAW The raw interrupt bit of the MB_HP_7_INT interrupt. (RO)

MB_HP_8_INT_RAW The raw interrupt bit of the MB_HP_8_INT interrupt. (RO)

MB_HP_9_INT_RAW The raw interrupt bit of the MB_HP_9_INT interrupt. (RO)

MB_HP_10_INT_RAW The raw interrupt bit of the MB_HP_10_INT interrupt. (RO)

MB_HP_11_INT_RAW The raw interrupt bit of the MB_HP_11_INT interrupt. (RO)

MB_HP_12_INT_RAW The raw interrupt bit of the MB_HP_12_INT interrupt. (RO)

MB_HP_13_INT_RAW The raw interrupt bit of the MB_HP_13_INT interrupt. (RO)

MB_HP_14_INT_RAW The raw interrupt bit of the MB_HP_14_INT interrupt. (RO)

MB_HP_15_INT_RAW The raw interrupt bit of the MB_HP_15_INT interrupt. (RO)

Register 22.23. MB_HP_INT_ST_REG (0x0054)

(reserved)																MB_HP_15_INT_ST																	
																MB_HP_14_INT_ST																	
																MB_HP_13_INT_ST																	
																MB_HP_12_INT_ST																	
																MB_HP_11_INT_ST																	
																MB_HP_10_INT_ST																	
																MB_HP_9_INT_ST																	
																MB_HP_8_INT_ST																	
																MB_HP_7_INT_ST																	
																MB_HP_6_INT_ST																	
																MB_HP_5_INT_ST																	
																MB_HP_4_INT_ST																	
																MB_HP_3_INT_ST																	
																MB_HP_2_INT_ST																	
																MB_HP_1_INT_ST																	
																MB_HP_0_INT_ST																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							

MB_HP_0_INT_ST The masked interrupt status bit of the MB_HP_0_INT interrupt. (RO)

MB_HP_1_INT_ST The masked interrupt status bit of the MB_HP_1_INT interrupt. (RO)

MB_HP_2_INT_ST The masked interrupt status bit of the MB_HP_2_INT interrupt. (RO)

MB_HP_3_INT_ST The masked interrupt status bit of the MB_HP_3_INT interrupt. (RO)

MB_HP_4_INT_ST The masked interrupt status bit of the MB_HP_4_INT interrupt. (RO)

MB_HP_5_INT_ST The masked interrupt status bit of the MB_HP_5_INT interrupt. (RO)

MB_HP_6_INT_ST The masked interrupt status bit of the MB_HP_6_INT interrupt. (RO)

MB_HP_7_INT_ST The masked interrupt status bit of the MB_HP_7_INT interrupt. (RO)

MB_HP_8_INT_ST The masked interrupt status bit of the MB_HP_8_INT interrupt. (RO)

MB_HP_9_INT_ST The masked interrupt status bit of the MB_HP_9_INT interrupt. (RO)

MB_HP_10_INT_ST The masked interrupt status bit of the MB_HP_10_INT interrupt. (RO)

MB_HP_11_INT_ST The masked interrupt status bit of the MB_HP_11_INT interrupt. (RO)

MB_HP_12_INT_ST The masked interrupt status bit of the MB_HP_12_INT interrupt. (RO)

MB_HP_13_INT_ST The masked interrupt status bit of the MB_HP_13_INT interrupt. (RO)

MB_HP_14_INT_ST The masked interrupt status bit of the MB_HP_14_INT interrupt. (RO)

MB_HP_15_INT_ST The masked interrupt status bit of the MB_HP_15_INT interrupt. (RO)

Register 22.24. MB_HP_INT_ENA_REG (0x0058)

(reserved)																MB_HP_15_INT_ENA MB_HP_14_INT_ENA MB_HP_13_INT_ENA MB_HP_12_INT_ENA MB_HP_11_INT_ENA MB_HP_10_INT_ENA MB_HP_9_INT_ENA MB_HP_8_INT_ENA MB_HP_7_INT_ENA MB_HP_6_INT_ENA MB_HP_5_INT_ENA MB_HP_4_INT_ENA MB_HP_3_INT_ENA MB_HP_2_INT_ENA MB_HP_1_INT_ENA MB_HP_0_INT_ENA																
																31																16
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	Reset		

MB_HP_0_INT_ENA Write 1 to enable the MB_HP_0_INT interrupt. (R/W)

MB_HP_1_INT_ENA Write 1 to enable the MB_HP_1_INT interrupt. (R/W)

MB_HP_2_INT_ENA Write 1 to enable the MB_HP_2_INT interrupt. (R/W)

MB_HP_3_INT_ENA Write 1 to enable the MB_HP_3_INT interrupt. (R/W)

MB_HP_4_INT_ENA Write 1 to enable the MB_HP_4_INT interrupt. (R/W)

MB_HP_5_INT_ENA Write 1 to enable the MB_HP_5_INT interrupt. (R/W)

MB_HP_6_INT_ENA Write 1 to enable the MB_HP_6_INT interrupt. (R/W)

MB_HP_7_INT_ENA Write 1 to enable the MB_HP_7_INT interrupt. (R/W)

MB_HP_8_INT_ENA Write 1 to enable the MB_HP_8_INT interrupt. (R/W)

MB_HP_9_INT_ENA Write 1 to enable the MB_HP_9_INT interrupt. (R/W)

MB_HP_10_INT_ENA Write 1 to enable the MB_HP_10_INT interrupt. (R/W)

MB_HP_11_INT_ENA Write 1 to enable the MB_HP_11_INT interrupt. (R/W)

MB_HP_12_INT_ENA Write 1 to enable the MB_HP_12_INT interrupt. (R/W)

MB_HP_13_INT_ENA Write 1 to enable the MB_HP_13_INT interrupt. (R/W)

MB_HP_14_INT_ENA Write 1 to enable the MB_HP_14_INT interrupt. (R/W)

MB_HP_15_INT_ENA Write 1 to enable the MB_HP_15_INT interrupt. (R/W)

Register 22.25. MB_HP_INT_CLR_REG (0x005C)

(reserved)																MB_HP_15_INT_CLR																	
																MB_HP_14_INT_CLR																	
																MB_HP_13_INT_CLR																	
																MB_HP_12_INT_CLR																	
																MB_HP_11_INT_CLR																	
																MB_HP_10_INT_CLR																	
																MB_HP_9_INT_CLR																	
																MB_HP_8_INT_CLR																	
																MB_HP_7_INT_CLR																	
																MB_HP_6_INT_CLR																	
																MB_HP_5_INT_CLR																	
																MB_HP_4_INT_CLR																	
																MB_HP_3_INT_CLR																	
																MB_HP_2_INT_CLR																	
																MB_HP_1_INT_CLR																	
																MB_HP_0_INT_CLR																	
31																16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	Reset
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0						

MB_HP_0_INT_CLR Write 1 to clear the MB_HP_0_INT interrups. (WO)

MB_HP_1_INT_CLR Write 1 to clear the MB_HP_1_INT interrups. (WO)

MB_HP_2_INT_CLR Write 1 to clear the MB_HP_2_INT interrups. (WO)

MB_HP_3_INT_CLR Write 1 to clear the MB_HP_3_INT interrups. (WO)

MB_HP_4_INT_CLR Write 1 to clear the MB_HP_4_INT interrups. (WO)

MB_HP_5_INT_CLR Write 1 to clear the MB_HP_5_INT interrups. (WO)

MB_HP_6_INT_CLR Write 1 to clear the MB_HP_6_INT interrups. (WO)

MB_HP_7_INT_CLR Write 1 to clear the MB_HP_7_INT interrups. (WO)

MB_HP_8_INT_CLR Write 1 to clear the MB_HP_8_INT interrups. (WO)

MB_HP_9_INT_CLR Write 1 to clear the MB_HP_9_INT interrups. (WO)

MB_HP_10_INT_CLR Write 1 to clear the MB_HP_10_INT interrups. (WO)

MB_HP_11_INT_CLR Write 1 to clear the MB_HP_11_INT interrups. (WO)

MB_HP_12_INT_CLR Write 1 to clear the MB_HP_12_INT interrups. (WO)

MB_HP_13_INT_CLR Write 1 to clear the MB_HP_13_INT interrups. (WO)

MB_HP_14_INT_CLR Write 1 to clear the MB_HP_14_INT interrups. (WO)

MB_HP_15_INT_CLR Write 1 to clear the MB_HP_15_INT interrups. (WO)